

Fenghao Liu

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Education

University of Michigan

M.S. IN ROBOTICS | GPA:4.0/4.0

Ann Arbor, MI, USA

Aug. 2025 - May. 2027

Southeast University

B.S. IN INFORMATION AND COMPUTING SCIENCE | GPA:3.53/4.0

Nanjing, China

Sep. 2021 - Jun. 2025

Experience

Neural Rendering & 3D Reconstruction (Accepted to OCEANS 2026)

Ann Arbor, MI, USA

RESEARCH ASSISTANT | FROG LAB, UNIVERSITY OF MICHIGAN

Mar. 2026 - Present

- **NeRF-Based 3D Perception with Sparse Sonar:** Built from scratch a SIREN/Lambertian implicit 3D baseline for extremely sparse, ill-posed side-scan sonar; evaluated NeuSSS, a NeRF-based sonar reconstruction method, and SIREN on real-world datasets.
- **Physical Model Tuning & Evaluation:** Analyzed underwater acoustic reflection characteristics to refine networks and physical rendering models through ablation studies; extracted 3D scene representations on simulated and real-world underwater datasets, providing a reliable benchmark for underwater robot spatial perception.

Tuniu (Xiao Niu Travel Assistant)

Nanjing, China

AI AGENT ALGORITHM INTERN

May 2025 - Aug. 2025

- **Multi-Agent Routing:** Designed a primary dispatch layer with Qwen3 and async verification mechanisms, achieving 93% routing correctness and an average latency of 2.57s.
- **Retrieval Optimization:** Deployed Milvus vector database and fine-tuned Sentence Transformer models, boosting retrieval precision by 2%.
- **Automated Evaluation System:** Developed automated factuality checks for POI data and itinerary feasibility, fixing critical data issues and enhancing the robustness of the production pipeline.

Selected Projects

WAM Precision Manipulation & DAGger Post-Training

Ann Arbor, MI, USA

GRADUATE INDEPENDENT RESEARCH | UNIVERSITY OF MICHIGAN

Aug. 2026 - Present

- **Full-Stack WAM Real-Robot Adaptation:** Adapted the pretrained 6.02B AHA-WAM (Video DiT-Action DiT) to a dual-camera SO-101 for precision cylinder and power-adaptor insertion; integrated multi-view observations, proprioceptive state/action interfaces, action chunking, and stateful visual memory for closed-loop inference.
- **Human-in-the-Loop DAGger Post-Training:** Built a takeover/handback correction loop and temporally structured dataset; supervised actions only under human control, retained pre-takeover context and eligible future-video targets, and optimized masked flow-matching objectives with 1:1 demonstration replay during full-parameter video-action co-training across 149 rollouts, 231 interventions, and 7,749 human-controlled frames.
- **Real-Robot Evaluation & Failure Recovery:** Ran controlled Segment-only/History/History+Future ablations and held-out video diagnostics; improved autonomous success from 7.7% to 46.2% on cylinder insertion and from 50.0% to 75.0% on power-adaptor insertion, identifying task-dependent effects of temporal context and predictive fidelity on recovery.

Closed-Loop Deployment & Offline DPO for Autoregressive VLMs (g0.5) on Real Manipulators

Ann Arbor, MI, USA

GRADUATE INDEPENDENT PROJECT | UNIVERSITY OF MICHIGAN

Jun. 2026 - July. 2026

- **End-to-End Pipeline & SFT Baseline:** Calibrated an SO-101 leader-follower setup and dual-camera LeRobot teleoperation pipeline; curated 156 demonstrations for 4-epoch full SFT on g0.5, then deployed low-latency WebSocket inference with action clipping and replanning.
- **Offline Trajectory Preferences:** Collected 56 policy rollouts/interventions across 8 edge-case initialization buckets; paired successes and failures within each task/bucket and unrolled trajectories into fixed-length windows for controlled preference learning.
- **Autoregressive Action-Space DPO:** Integrated the GRAPE/TPO formulation at the g0.5 action-token level with a joint DPO + Chosen CE objective; eliminated dual-model OOM during full fine-tuning by precomputing and caching reference log-probabilities, improving convergence stability over pure SFT within 1 epoch.

Fine-Tuning & Ablation Analysis of Diffusion VLA Models ($\pi_{0.5}$)

Ann Arbor, MI, USA

GRADUATE INDEPENDENT PROJECT | UNIVERSITY OF MICHIGAN

Jun. 2026

- **Pipeline Reproduction & Efficient Fine-Tuning:** Reproduced the $\pi_{0.5}$ underlying training pipeline on 4x A100 (40GB) GPUs using the OpenPI/JAX/Flax framework, conducting large-scale multi-modal fine-tuning experiments on the LIBERO-Spatial simulation dataset.
- **Modular Fine-Tuning & Ablation Analysis:** Compared the performance of full fine-tuning, PaliGemma LoRA, and Action Expert LoRA; quantitatively analyzed the root cause of suboptimal performance in pure action-expert fine-tuning, validating that the spatial reasoning capability of VLA models is highly coupled with the underlying logic of vision-language foundation models (e.g., SigLIP/PaliGemma), providing crucial theoretical support for real-world model selection.

5-DoF Manipulator Kinematics & Vision-Guided Control

Ann Arbor, MI, USA

GRADUATE PROJECT | UNIVERSITY OF MICHIGAN

Aug. 2025 - Oct. 2025

- **Kinematics & Perception-to-Control Loop:** Implemented PoX forward kinematics (1.9% mean error) and geometric IK; calibrated camera extrinsics via AprilTag PnP/homography and integrated HSV localization, IK, and joint commands into a "Click-to-Grab" sorting system.

Skills

Programming & Systems	Python, C++, Linux, Git, Docker, CMake, WebSocket, OpenCV, Hugging Face
Robot Learning	PyTorch, JAX/Flax, BC/HG-DAgger, DPO, VLA/WAM, NeRF, Flow Matching, LeRobot/OpenPI, Multi-GPU/DeepSpeed
Robotics & Control	Manipulation, Teleoperation, PoX/IK, Camera Calibration, Motion Planning, PID Control, ROS 2